AI Automated Podcast Transcript-Medi Lens

End-to-end AI system for podcast transcription and clinical analysis

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PROBLEM STATEMENT

Transforming Medical Audio Into Actionable Intelligence

Healthcare organizations face significant challenges in managing and analyzing audio content from clinical encounters, medical podcasts, and patient consultations.

Manual transcription is time-consuming, expensive, and prone to errors. Extracting meaningful insights, clinical summaries, and trending topics requires substantial human effort.

Key Challenges

- *Time-intensive manual transcription processes*
- *Difficulty extracting clinical insights at scale*
- *Limited ability to analyze sentiment and themes*
- *No standardized workflow for medical audio analysis*

PROJECT GOALS

Core Objectives

Accurate Transcription

Convert medical audio into precise, timestamped transcripts automatically using advanced ASR technology

Sentiment Analysis

Analyze emotional tone and sentiment across each audio segment for deeper understanding

Keyword Extraction

Identify and visualize key medical terms, topics, and concepts through intelligent keyword clouds

Clinical Summaries

Generate automated, structured clinical summaries from medical dataset audio

Interactive Interface

Provide intuitive UI for uploading audio files and navigating segmented content seamlessly

Methodology

Audio Upload & Preprocessing

Audio files processed and optimized for transcription

ASR Transcription

OpenAI Whisper converts speech to accurate text

Topic Segmentation

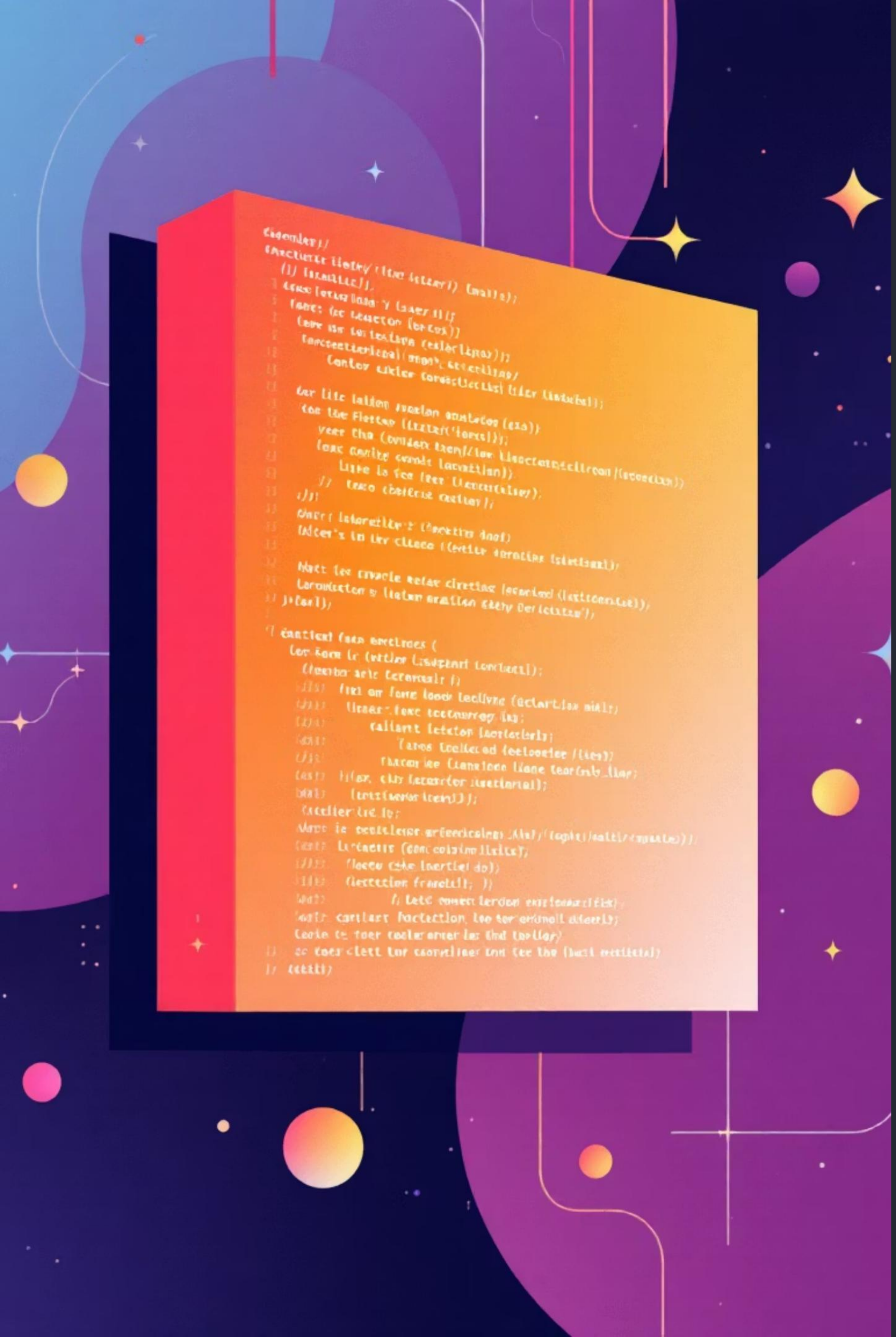
Content divided into logical thematic segments

NLP Analysis

Sentiment, keywords, and clinical insights extracted

Output Generation

Summaries, visualizations, and interactive UI delivered



Tech Stack

Core Technologies

- ***Python*** *- Primary language*
- ***Librosa & PyDub*** *- Audio processing*
- ***OpenAI Whisper*** *- Speech-to-text ASR*

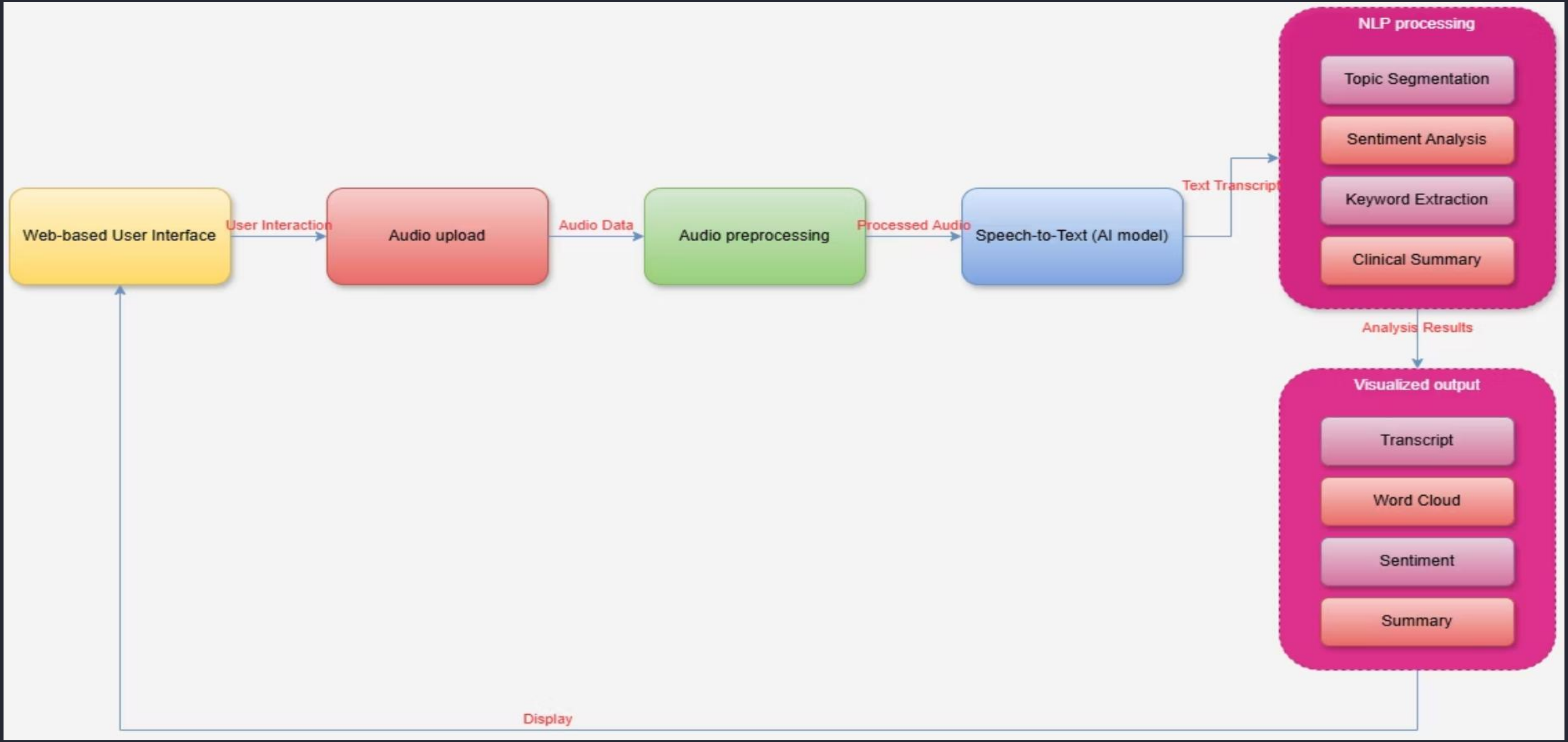
NLP & Analysis

- ***NLTK & SpaCy*** *- Text processing*
- ***Hugging Face*** *- Transformer models*
- ***TF-IDF*** *- Keyword extraction*

Visualization & UI

- ***Streamlit*** *- Interactive interface*
- ***Plotly*** *- Data visualization*
- ***WordCloud*** *- Keyword visualization*

System Architecture



Key Features & Capabilities

Automatic Audio Transcription

High-accuracy speech-to-text conversion with timestamp precision for medical terminology and clinical language

Segment-Level Sentiment Analysis

Granular emotional tone detection across conversation segments to identify patient concerns and clinical interactions

Keyword Cloud Visualization

Dynamic visual representation of key medical terms, conditions, and topics for rapid content understanding

Clinical Summary Generation

AI-powered summarization of medical encounters with structured output following clinical documentation standards

Interactive Streamlit UI

User-friendly interface for seamless audio upload, transcript navigation, and insight exploration

INNOVATION ROADMAP

Future Scope & Enhancements

1

Real-Time Transcription

Extend capabilities to process live podcast streams and clinical conversations with minimal latency

2

Multi-Language Support

Add support for multiple languages to serve diverse healthcare settings and global medical content

3

Domain-Specific Models

Integrate specialized medical language models to improve clinical summary accuracy and terminology recognition

4

Emotion Recognition

Incorporate advanced speech emotion recognition to capture nuanced patient and provider interactions

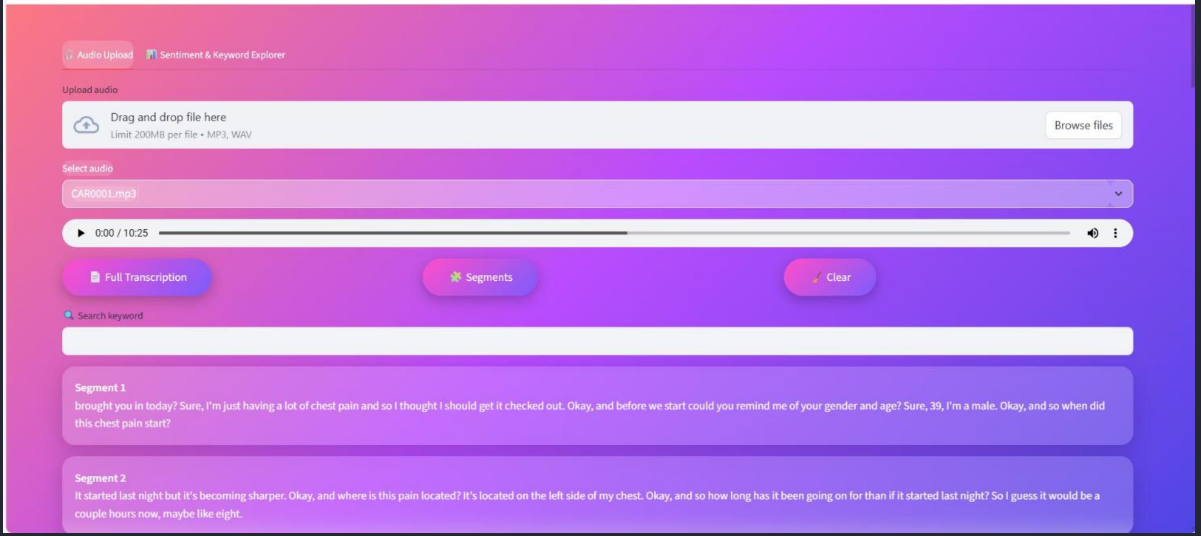
5

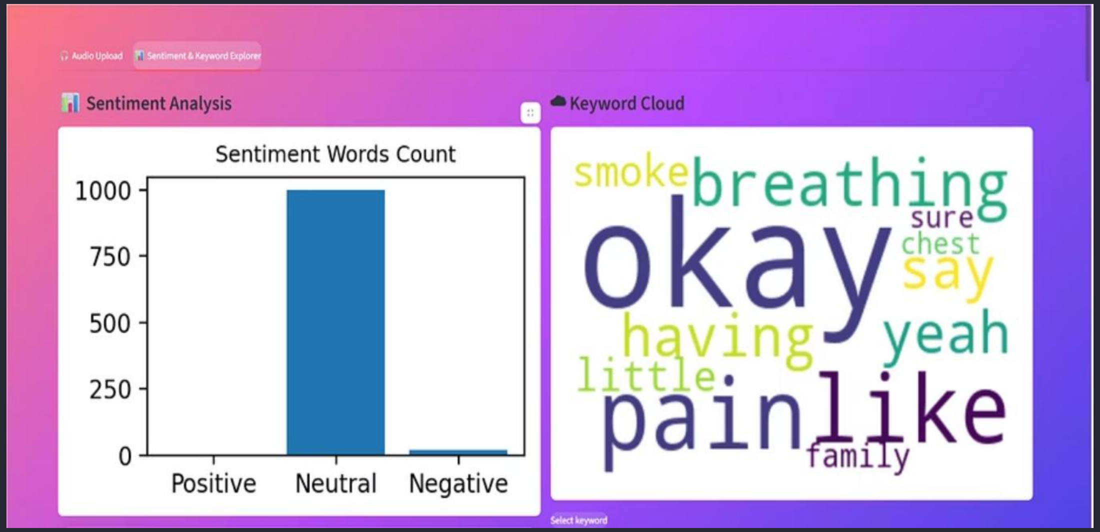
Analytics Dashboard

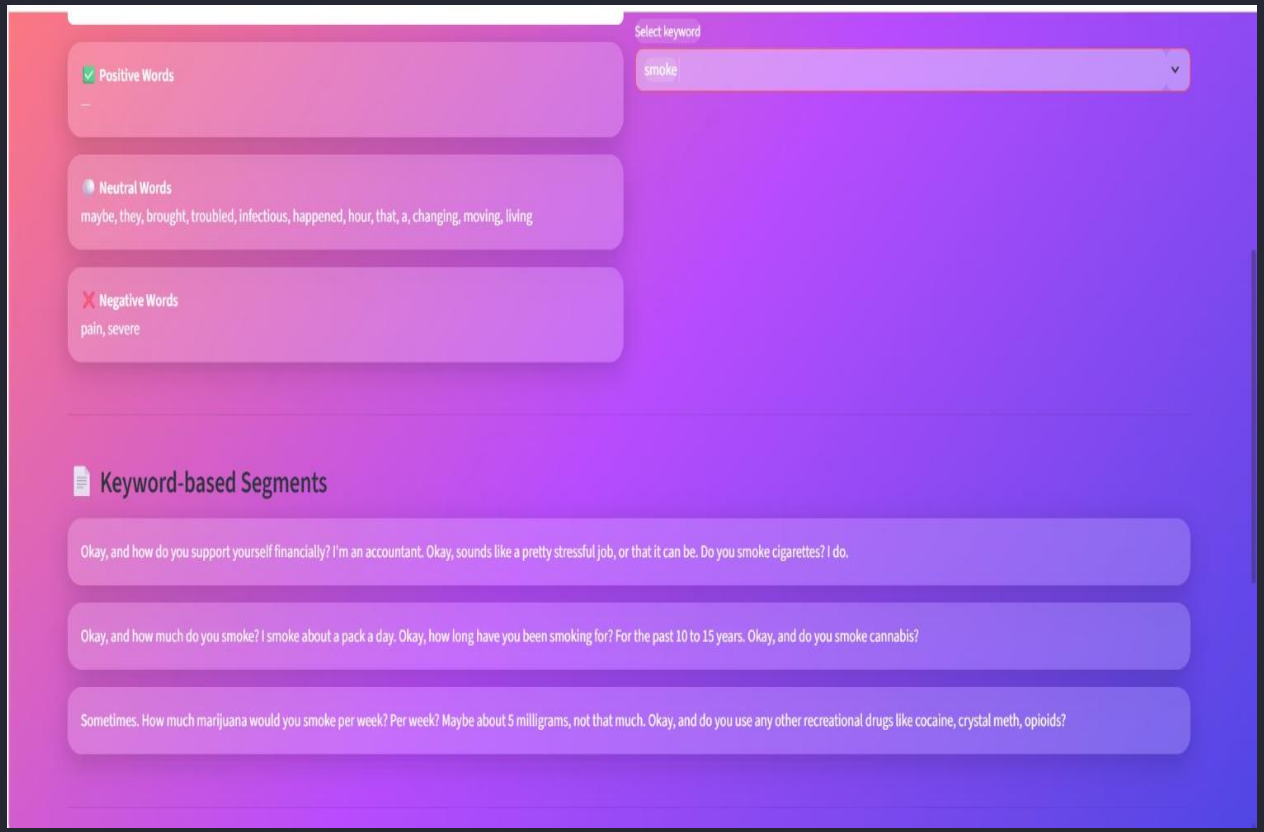
Build comprehensive dashboard for analyzing trends across large medical audio datasets with reporting features

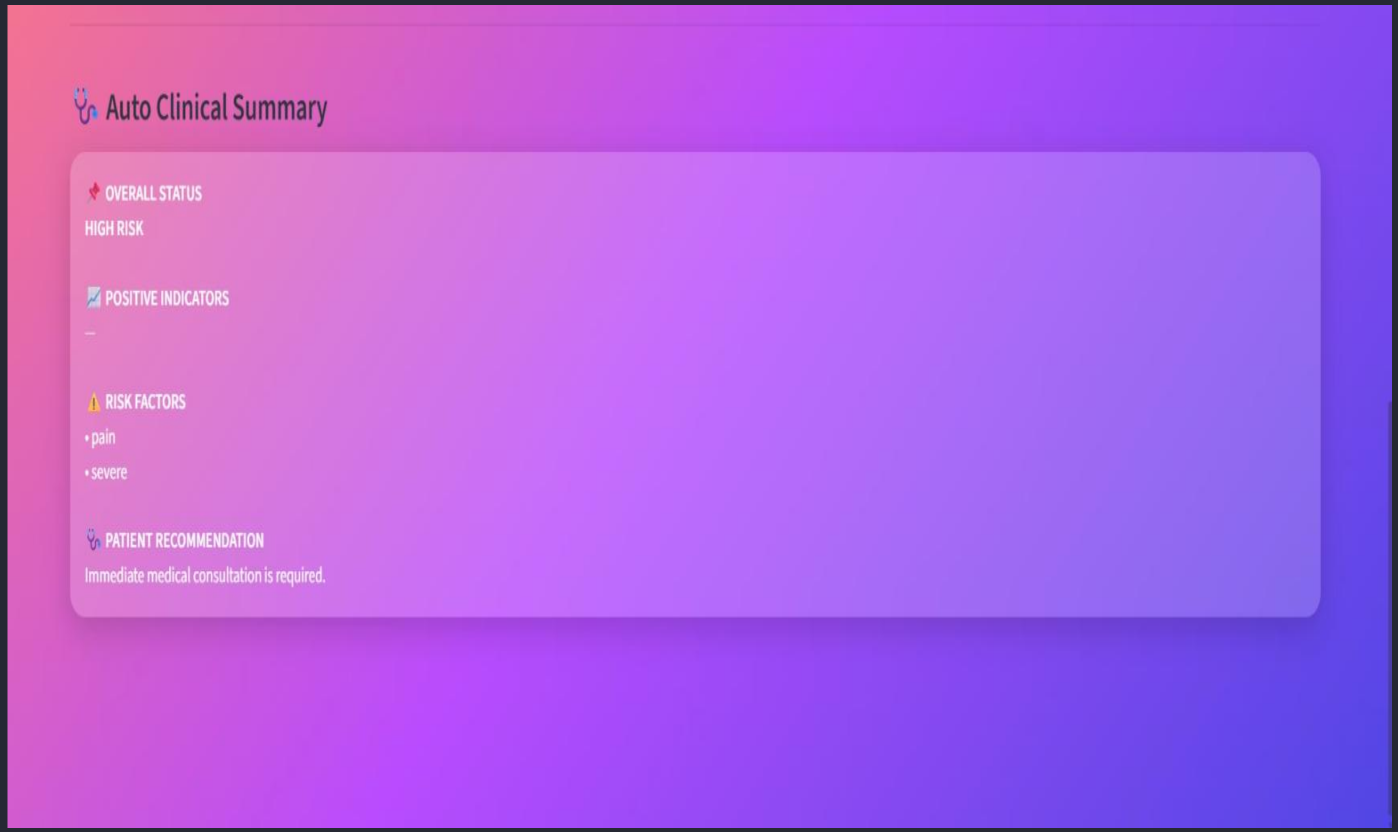
System Demonstrations

Interactive workflow showcasing the complete transcription and analysis pipeline






Select keyword
smoke
Positive Words
—
Neutral Words
maybe, they, brought, troubled, infectious, happened, hour, that, a, changing, moving, living
Negative Words
pain, severe
Keyword-based Segments
Okay, and how do you support yourself financially? I'm an accountant. Okay, sounds like a pretty stressful job, or that it can be. Do you smoke cigarettes? I do.
Okay, and how much do you smoke? I smoke about a pack a day. Okay, how long have you been smoking for? For the past 10 to 15 years. Okay, and do you smoke cannabis?
Sometimes. How much marijuana would you smoke per week? Per week? Maybe about 5 milligrams, not that much. Okay, and do you use any other recreational drugs like cocaine, crystal meth, opioids?


Auto Clinical Summary
OVERALL STATUS
HIGH RISK
POSITIVE INDICATORS
—
RISK FACTORS
• pain
• severe
PATIENT RECOMMENDATION
Immediate medical consultation is required.

Technical References & Resources

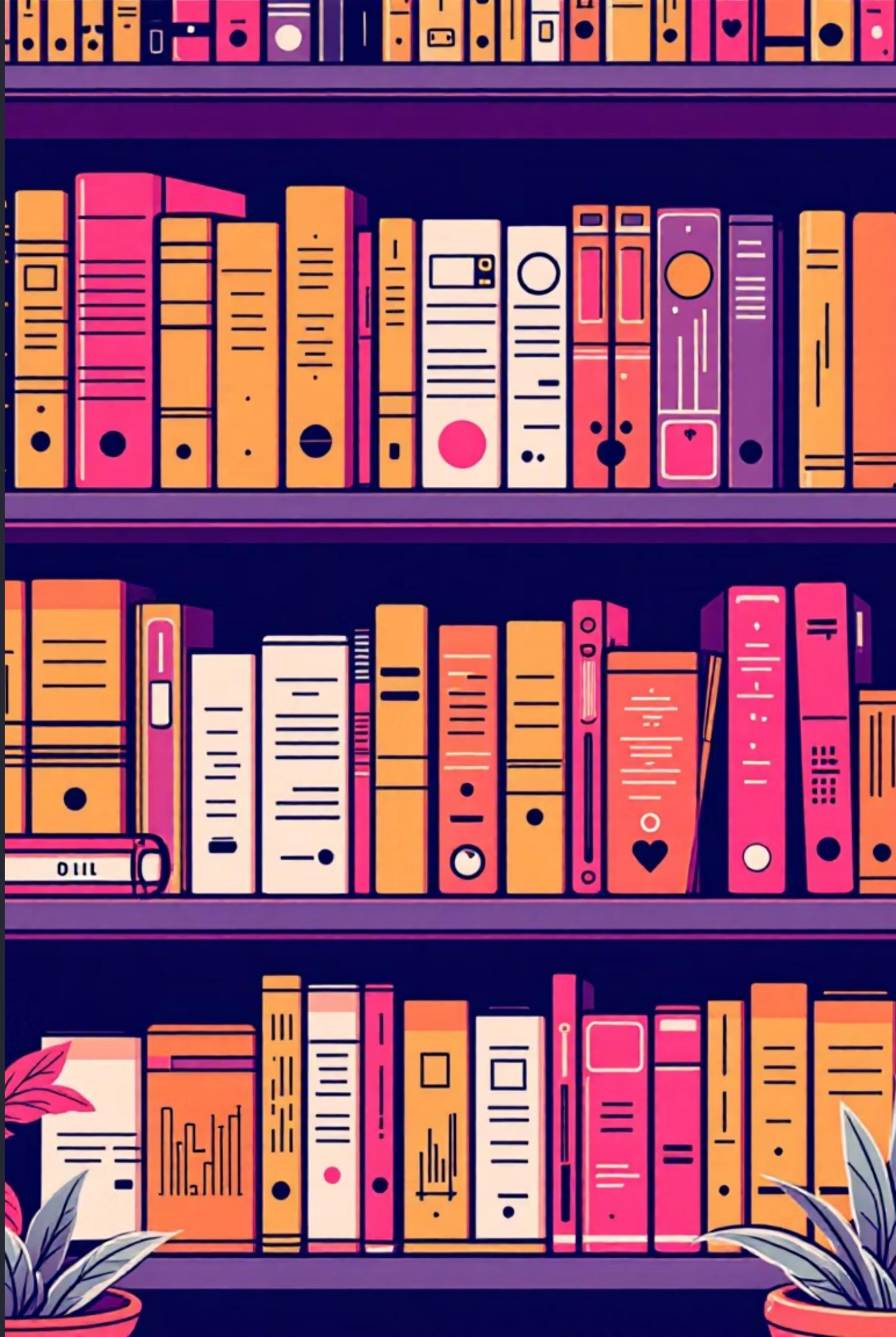
Core Libraries

- *OpenAI Whisper - ASR transcription*
- *Librosa - Audio analysis*
- *NLTK - Natural language toolkit*
- *Scikit-learn - Machine learning*

Visualization & UI

- *Streamlit - Interactive UI framework*
- *WordCloud - Keyword visualization*
- *Matplotlib - Plotting library*

Academic Reference: *Speech & Language Processing by Jurafsky & Martin - Stanford NLP Resource*



Q&A

Thank You